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Internet of Things-Based Middleware Against Cyber-Attacks on Smart Homes using Software-Defined Networking and Deep Learning

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***Abstract*— Internet of Things (IoT) devices are expected to number about 3.5 billion by 2023.; a tremendous amount of Internet of Things (IoT) data that is generating (IoT) devices is estimated to exceed 79.4 zettabytes by 2025. Security challenges will become an increasingly significant issue, especially in smart homes that depend entirely on IoT devices. Due to the weak infrastructure of the IoT, it is vulnerable to various types of cyber-attacks. The most common IoT attacks are distributed denial of service (DDoS). Most traditional security solutions, like intrusion detection systems (IDS), cannot detect most attacks. Complexity is being hidden by a new paradigm that recently arose, called the software-defined system that is brings a significant change to the networking industry, a great solution for mitigation of attacks that can adopt deep learning technique to encounter cyber-attacks based on the attack behavior and by filtering normal and attack traffic by using well-defined rules. This paper proposed a system by suggesting middleware that can help mitigate or prevent various attacks on IoT on smart home environment. Machine learning has included in the middleware to provide automatic protection against cyber-attacks on IoT networks. A promising approach to protecting real-time, highly accurate attacks on SDN-managed IoT networks has been proposed. This middleware allows IoT devices to efficiently handle evolving security threats dynamically and adaptively without impacting the IoT devices.**

Keywords*— *IoT Security, Smart Home, Software-Defined Networking, Cyberattack, Middleware

I. INTRODUCTION

The Internet of Things encompasses various devices ranging from cameras, wearable, Smart factory equipment, household appliances to medical gadgets, automobiles, and even nuclear reactors, due to its ability to transfer data across a network to the virtual world[1]. The Infrastructure for the Internet of things is lack because it a small device with limited capacity, hardware, battery life, incapable of applying security procedures, and The nature of communication may lack confidentiality and authenticity, leaving it vulnerable to attack like Botnets, ,distributed denial-of-service (DDoS), Eavesdropping, Man-in-the-Middle, Firmware Hijacking, spoofing and Brute Force Password Attack[2]. The great growth of IoT and its huge traffic is critical to implement security measures to counter these types of attacks. Emerging techniques like Software-defined networking (SDN) and Deep learning capable of building intelligence systems to analyze the huge traffic of data and mitigate, prevent cyber-attack.

Software-Defined Networking is a new intelligent, centralize, and dynamic network architecture that helps device management[3]. Whatever of the network technology is used to connect the devices, this was accomplished by separation of control, data and application plane that brought more security for IoT devices. by using SDN based on An OpenFlow switch for IoT devices for and traffic analysis at the northbound and southbound to find the malicious attack from that's traffic and act before escalated.

The application of machine learning is fundamentally altering practically every technical field ranging from pattern recognition for autonomous vehicles to Network security[4]. Deep learning is an intelligence technique to face Cyber Security Threats, e.g., filtering massive traffic of data in the network and recognize the harmful pattern or malicious data traffic [5]. This is a tremendous asset when fighting against numerous cyber-attacks, such as SQL injection or DDOS attacks, etc.

This paper presents a middleware for real-time cyber-attack detection and prevention that provide solutions for Smart Homes by integrating deep learning with SDN, which aims to provide self-protection in a traffic Internet of Things network in the smart home environment; that method depends on the classification of extracted features for IoT data traffic.

Aims of this paper summarized are as follows:

A- We Proposed a middleware for a smart home with low cost using Raspberry Pi and Zodiac-Fx SDN open flow Switch.

B- We Suggested that middleware architecture to be integrated with Deep learning techniques that prove the proposed system by implementing deep learning that provides a great performance.

The paper structured as follows: The Section II discussed the related work in IoT Security and Threats, IoT Security based on SDN and The existing Machine learning applications for IoT security. Section III provides a concise overview of the SDN architecture. Section IV describes the architecture of our proposed middleware against cyber-attacks with components and Machine learning with deep learning

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implementation. Section V described the results and discussions that show the performance evaluation of the deep learning model—section VI Conclusion with a plan.

II. Related Work

The growth of the Internet of things devices massive, On the other hand, vulnerabilities on those devices and slight protocols that can exploit, lead to organized cyberattacks and breaches, private user information accessibility, or the ability to take advantage of users' devices in the future actions, Urgent need to develop secure systems have become increasingly crucial and ever more important against the cyber-attacks. The cyberattacks can be performed on various layers on the IoT platform. Perception layer Unauthorized access, Spoofing, and Eavesdropping attack, Network layer, e.g., Sinkhole, DDoS MitM attack, e.g., on Middleware layer Insider attack Application layer Sniffing and Spear-Phishing attack, In this section we summarized the studies and work that closed to our proposed system.

A. IoT Security and Threats

Each IoT platform and device has a different set of capabilities, and these capabilities will determine what type of security solutions are required. Jarkko ,et al [6]

Proposed security system for IoT that can Mitigate attack using a trusted Network Edge Device (NED) with a connection of the lot gateway via a LAN cable and apply security policies monitor IoT connection with control via NED. By Caiming et al. [7] The biological immunology method has been proposed for IoT environment the self and detector antigen and the immunity-based antigen are simulated which consist of sensor devise, IoT gateway and server that execute The proposed approach . The approach is based on cloning attack, and modify it, then perform replay attacks and modify that replay attacks. Seul et al. [8] have proposed a paradigm that goal to counter security threats in IoT devices by checking the state of IoT devices on a continuous basis to discover early indicators of irregularity that system based on a web service.

B. IoT Security based on SDN

SDN is a good solution to increase the performance of IoT and provide robust security.

Suman Bhunia et, al.[9] presented a framework called Soft Things, which uses SDN for IoT and detection of aberrant behavior and attacks, to try to secure the Internet of Things (IoT) and guard against attacks and suspicious activity before they happen, The framework is capable of detecting attacks on the Internet of Things with roughly 98% accuracy. Kallol et al. [10] developed a security architecture around SDN for IoT networks, allowing authentication and authorization of IoT devices using a lightweight protocol and dynamic policy-driven. used ONOS Controller and Raspbian Virtual Machines for architecture validation.[11] In order to counter attacks on IoT, the authors created a system model that incorporates SDN and IoT networks. Implemented on, Kodi Media Center using Raspberry Pi, and OpenFlow Protocol, the intended system model was to provide solutions for mitigating cyber-attacks against IoT [12]. Kingshuk and Chaitanya proposed an approach that used SDN and Edge Computing to supply security to IoT. This is a hypothetical situation considering two different approaches that are taken with network access endpoint devices and Cellular Mobile Operators. [13] A preliminary study called Cluster SDN has been presented by Carlos et al. To make a simulated system that consists of 500 devices managed by SDN, the project uses network virtualization and OpenFlow to create virtual nodes. In the results, the controller-set network devices can only forward packets according to established rules and control an IP header on a higher level of the application.[14] Yaser et al , proposed Middleware architecture for IoT administration process using “A Software Defined based Internet of Things framework that consists of the three-layers physical layer, Middleware layer and Application layer”, The physical layer consists of Sensor Network (SN) to collected data and database pool cluster work as data warehouse and data related to configuration. The control layer contains IoT, SDN and SDStore controllers. Application layer which facilitates end users' access and use of stored data.

C. IoT Security based on Machine learning

Machine-learning and Blockchain have recently been used to address issues in IoT security The IoT device classification framework proposed by [15], [16] authors analyses HTTP packets for classification purposes. use this approach to first separate IoT devices from non-IoT devices, and then classifiers are used to classify into nine different types of IoT devices [17]. Yair et al, have used a machine learning algorithm random forest for extracting features from IoT network traffic to identify reliable IoT from the prior connected device from the whitelist, The collected dataset consists of 17 features the overall performance was 96% of Accuracy. Baibhab and Shovan [18] Proposed Radio-frequency Physical unclonable functions (RFPUF) deep learning framework for utilising inherent process variation in RF properties to authenticate wireless nodes in real-time of Transmit and Receive (RX and Tx) the experiments implemented on 4800 transmitters and overall accuracy 99.9 %. Francisco et al. [19] used machine learning- for DoS detection from IoT network traffic data. The model has been performed on a publicly dataset, the Highest accuracy for Random Forest classifier (RF) with an accuracy of 99.6%.

III. SDN Background

A. Software-Defined Networks (SDN)

This innovative paradigm called the software-defined systems has overcome all the complexity of traditional Network system architecture., A key function of an SDN It allows network administrators to dynamically change the logical network topology in real-time. SDN has typically been applied to create, implement, and administer network infrastructures that separate the control from data planes in order to provide speed, security and centralized intelligent for a better user experiences Architecture consist of Three layers [20] Data Plane, Control Plane and application layer and two bridge API Northbound and Southbound their function to initialize connection between the previous layers.

- Data Plane it takes place through a south-bound API and contain Network devices physical or Virtual. it responsible only for Forwarding data in accordance with set policies Which provides an efficient Forwarding mechanism.

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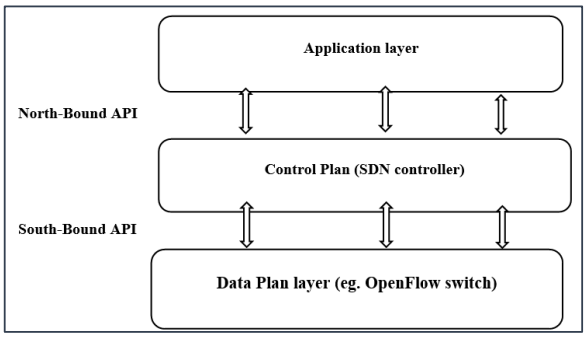


Fig. 1. SDN Architecture

- Controller plane. SDN controller is a smart 'brain' of the network due manages all networking devices and increase the global utilization of resources and improve the efficiency of resource management ,2 [21] using the OpenFlow Protocol at the beginning by applying predefined rules, collects the switches' statuses, and uses rules to control the switches' entries (flow rules Table). When undefined events occur, the controller then handles requests. The controller connects with the data plan via OpenFlow protocol that exchanges the message between the controller and switch that supported OpenFlow, the pure OpenFlow switch contains an operating system only for boot. and the flow table is written and configure by The SDN controller, An OpenFlow table consists of three parts: Statistics features, match fields and actions as shown in Fig. 2 that analyses packets to determine which flow tables that consist of contains flow entries, and packets then match. Packet based on the precedence of flow entries.

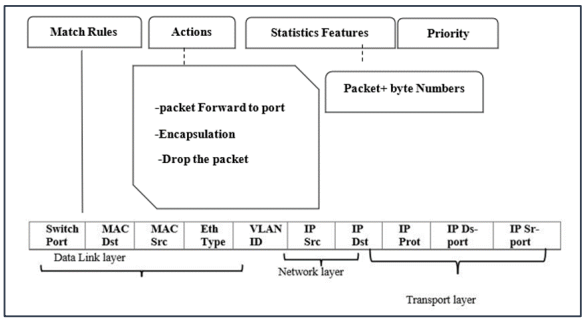


Fig. 2. OpenFlow flow- table Entries.

- The application layer comprises all applications that perform various functions. It is an open layer to innovate and utilize network information in terms of network architecture, network statistics ,network state , etc., on the application layer. The main applications for, network configuration, network automation, network management are network configuration and management, network automation load balancing, etc.

IV. Proposed System

A business enabler for the Internet of Things (IoT) is emerging quickly, but it is also bringing about new challenges for network and security teams alike. Due to a large amount of data stored and retrieved securely, IoT devices have a limited amount of memory and computation time. Therefore, it is necessary to have a dynamic framework that overcomes the previous security and storages issues in traditional network infrastructure, The new paradigm of insecure applications and IoT services that is being implemented by software-defined networking (SDN) gives network administrators complete control over network behavior and prevents network load balancing, We introduced SDN with Machine learning to overcome and mitigate cyber-attacks based on Middleware solution, which will be deployed on the Internet of Things Gateway and SDN application layer.

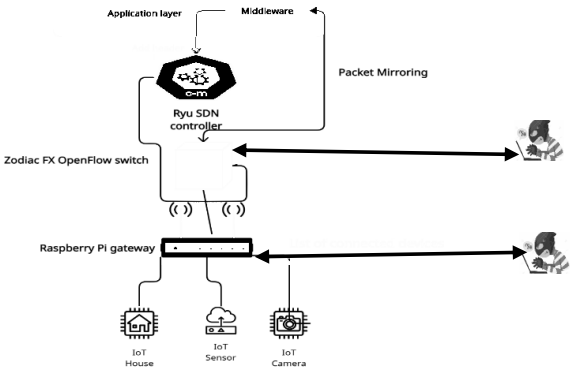


Fig. 3. Proposed Model.

A. System components:

In our preliminary study for the proposed system, we suggest using the Zodiac-Fx switch support OpenFlow protocol developed by Northbound Networks to build real-time SDN applications with low cost for research purposes that can be used for smart home security [22]. It features four available ports and has built-in support for OpenFlow v1.3, it includes open-source firmware, and support till 512 entry flow table.

B. SDN controller

Mid-layer (control Plan) controllers are present. SDN Controllers, such as Ryu, ONOS, Open Day light Controller. Ryu controller was uniquely designed for high-traffic rates and network agility. APIs are integrated into Ryu along with well-defined software components. Having the developers create an application and maintaining various networking devices. Python is used to write the code for the Ryu controller [23]. We suggest Using Ryu controller and install on Raspberry pi 4 with Low-cost

" $ sudo apt-get install gcc python-dev libffi-dev libssl-dev libxml2-dev libxslt1-dev zlib1g-dev python-pip"

command install ryu

"$ sudo pip install ryu"

C. Raspberry Pi Gateway

The Raspberry Pi is a small computer size and runs a full Linux operating system [24]. The CPU and GPU are integrated into a single integrated circuit with the rest of the RAM, USB ports, and other internal components adhered onto the board. And there are many versions of Raspberry Pi with superlative features in our system we suggest raspberry pi 4 with 8 GB RAM As the gateway for IoT devices that will connect using and mode Bluetooth, ZigBee, RFID and NFC, Wi-Fi.

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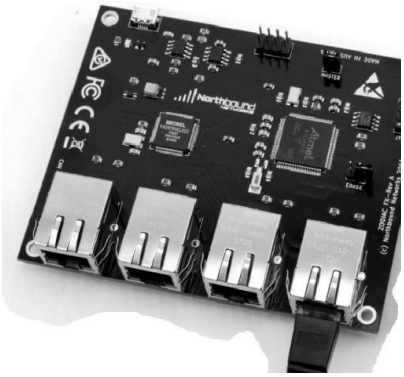


Fig. 4. Zodiac-Fx SDN open flow Switch..

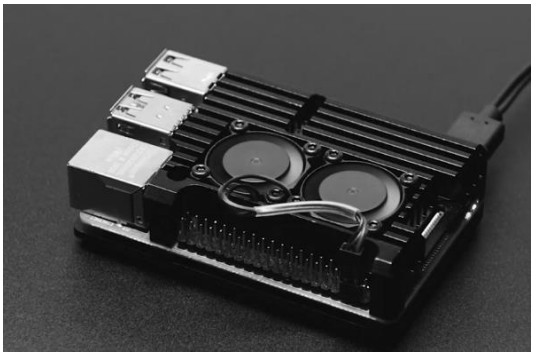


Fig. 5. Raspberry Pi 4 computer.

D. Middleware for attack Detection

our proposed Middleware that will be used on the SDN for Attack detection and mitigation based on data traffic of IoT network by utilizing Machine learning and deep learning, with smart home-based IoT devices hitting the market quickly, their security could be lacking. Privacy issues aren't addressed in some cases, making it difficult to be sure your device is secure. Hackers have been able to view what is going on inside a house using baby monitors and security cameras that have been hacked. These threats were the impetus to create a secure middleware to protect smart homes at a lower cost.

On south-bound API, the data Traffics is always distributed across OpenFlow switches. The OpenFlow switch checks the packet's information, like source IP address and source, destination port; when a packet arrives, the switch compares the incoming packet with all the active flow entries. Once it finds a match, the switch executes the appropriate action [25]. The packet is forwarded to the controller for further instructions if no match is found. In our proposed system, we suggest using switch supports OpenFlow with minimum cost, as shown in Fig. 4 with details, that OpenFlow switch directly with Raspberry pi 4 that is used as a gateway for IoT Devices, and sensors or actuators as shown in Fig. 3.

Data collection for middleware on Northbound and procedures as the following:

1-Input: incoming packet

2-While Packet arrived at switch Do:

3-Middleware sends a request command for SDN to collect data.

4-Then Mirror copy for arrived packet

5-send it to middleware

6- classify packet

7-If packet abnormal:

8-Add header (1) to the packet if abnormal

9- send it to the controller with priority

10-Output: SDN takes drop action

11-End if

The traffic data can collect for Middleware through the OpenFlow protocol from the Zodiac-Fx port, The middleware sends state command "Sudo ovs-ofctl dump-flows s(switch number)" to SDN for starting data traffic collection, there is no pledge all features that they include can retrieve in open flow switch in real-time so we suggest packet mirroring to middleware for classification and overcome latency of forwarding packet to the destination before verifying in sdn controller ,In middleware, the incoming packet feed to a deep neural network classifier, from packet behavior if the packet is malicious add header to packet as a flag (1) using P4 (programming language) and forwarded it to SDN to take drop action if packet normal left and drop in middleware. And open flow switch continues to forward the packet to the destination.

Data	Switch Port	MAC Src	MAC Dst	Eth Type	VLAN ID	IP Src	IP Dst	IP Prot	IP Sr-port	IP Ds-port	1

Fig. 6. Data and Flow table structure with Flag for Abnormal Packet.

To ensure the system is securable for both Inbound as well as Outbound Thus making sure that the devices are safe from any internal as well as an external attack, from prior study proposed by Mahmoud, et al "Management Platform (IoT-MP) [26]", so we suggest tiny middleware in the gateway for IoT-connected devices to provide authentication, it contains a unique id for each device the name and others that would not allow for unknown devices to connect to the IoT platforms.

E. Implementation Machine learning classifier

Our Random Forest classifier has been applied on one type of attack DDos Attack as shown in Fig. 7, model consists of four phases to collect data for training that consists of six features as shown in Table I, Real-time data collection from open flow switch, Feature extraction, and Training model and classification.

The train Data was collected from open flow mininet traffic using Wireshark using the command “sudo Wireshark &>/dev/null &”. The topology consisted of 4 open flow switches and eight hosts, as shown in table I. The DDoS features collected using hping3 script that performs attack using command “sudo h4 hping3-faster -- rand-source -I u50000 -c 2000 10.0.0.5” host4 performed an attack on host3 IP 10.0.0.5.

The features related to DDos Attack extracted from the open-flow switch using SDN Controller that sends flow sate message to open flow switch, and it sends the flow table.

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After extracting features related to DDoS attacks, as described in table I, these featured feed to a trained model (RF-Classifier) to identify the malicious traffic from normal traffic. If traffic is abnormal select the attack host or switch and block the port. The experiment applied on two types of attack UDP flood, TCP SYN flood attack usually if its value greater than 30000 that's mean there is attack, The overall accuracy of the RF-Classifier for each type of attack is 92.2 % For UDP flood and 93.1 for TCP SYN flood.

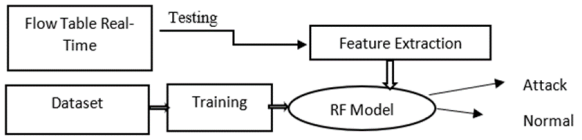


Fig. 7. Random Forest Attack detection.

TABLE I. EXTRAXTED FEATURES

No	Features
1	SFE (Speed of Flow Entry)
2	SSIP (Speed of Src IP)
3	RFIP (Ration of Flow Pair)
4	SDFB (Std deviation of flow bytes)
5	SDFP (Std deviation of flow packets)

F. Deep learning Experiment.

Middleware based on the DL technique has been proposed for detecting attack networks in an IoT-based environment. The characteristics of deep learning feature learning and classification with high performance and Accuracy. We have implemented a deep learning model as a good experience and proof for our proposed system. The newest dataset, called IoTID20, collected by Imtiaz, Qusay [27], contains 86 features; after reducing features, it became 81 features. We performed pre-processing by removing null, malformed values, labeling using label encoder, scaling them using a standard scaler, and data splitting using the traditional method 70% for training and 20% for testing, in the second experiment using K-fold to 10 groups to overcome the overfitting.

1) Deep learning Technique.

The experiment implemented on Convolutional Neural Network (CNN) with Long Short-Term Memory networks (LSTM) for classification cyber-attack one time on two classes Normal or abnormal and second experiment on five classes Mirai, DDos, scan cat, Normal and MITM ARP Spoofing, as shown in Fig. 8 the features that have been used were 81 as shown in table II.

TABLE II. THE REDUCED FEATURES

Feature name	
Protocol	FIN_Flag_Cnt
Flow_Duration	SYN_Flag_Cnt
Bwd_Pkt_Len_Std	RST_Flag_Cnt
Flow_Byts/s	PSH_Flag_Cnt
Flow_Pkts/s	ACK_Flag_Cnt
Flow_IAT_Mean	URG_Flag_Cnt
Flow_IAT_Std	CWE_Flag_Count
Flow_IAT_Max	ECE_Flag_Cnt
Flow_IAT_Min	Down/Up_Ratio
Fwd_IAT_Tot	Pkt_Size_Avg
Fwd_IAT_Mean	Fwd_Seg_Size_Avg
Fwd_IAT_Std	Bwd_Seg_Size_Avg
Fwd_IAT_Max	Fwd_Byts/b_Avg
Fwd_IAT_Min	Fwd_Pkts/b_Avg
Bwd_IAT_Tot	Fwd_Blk_Rate_Avg
Bwd_IAT_Mean	Bwd_Byts/b_Avg
Bwd_IAT_Std	TotLen_Bwd_Pkts
Bwd_IAT_Max	Fwd_Pkt_Len_Max
Bwd_IAT_Min	Fwd_Pkt_Len_Min
Fwd_PSH_Flags	Fwd_Pkt_Len_Mean
Bwd_PSH_Flags	Fwd_Pkt_Len_Std
Fwd_URG_Flags	Bwd_Pkt_Len_Max
Bwd_URG_Flags	Bwd_Pkts/b_Avg
Fwd_Header_Len	Bwd_Blk_Rate_Avg
Bwd_Header_Len	Subflow_Fwd_Pkts
Fwd_Pkts/s	Subflow_Fwd_Byts
Tot_Fwd_Pkts	Subflow_Bwd_Pkts
Tot_Bwd_Pkts	Subflow_Bwd_Byts
TotLen_Fwd_Pkts	Fwd_Seg_Size_Min
Bwd_Pkt_Len_Min	Active_Mean
Bwd_Pkt_Len_Mean	Active_Std
Bwd_Pkts/s	Active_Max
Pkt_Len_Min	Active_Min
Pkt_Len_Max	Idle_Mean
Pkt_Len_Mean	Idle_Std
Pkt_Len_Std	Idle_Max
Pkt_Len_Var	Idle_Min
Init_Fwd_Win_Byts	Label
Init_Bwd_Win_Byts	Cat
Fwd_Act_Data_Pkts	

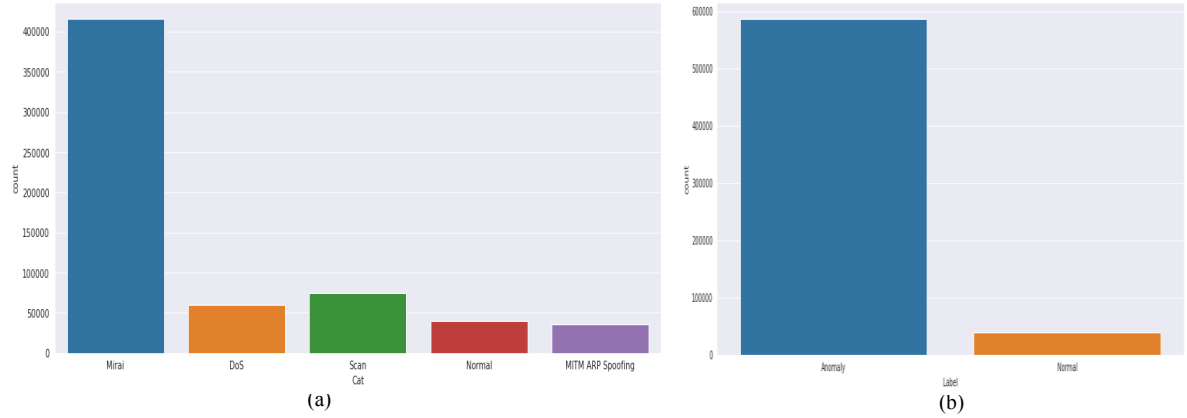


Fig. 8. (a) multiclasses ,(b) Two classes.

We set the LSTM and CNN models with Adam, Stochastic gradient descent (SGD), and Rmsprop optimizers, then iteration for 100 epochs. Binary classification and Multi-class classification we used the same parameter. Once the models

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were built, we began training them over the parameters that were set out. At the end of each epoch, we determined how well the model had performed and saved the model weights if the model gives the best accuracy. The evaluation processes. to identify whether the best classifiers for binary or multi-class.

V. Results And Discussion

The experiments in this paper are conducted on the Mininet virtual open flow switch and hosts emulator and Ryu Sdn controller install on Ubuntu in Raspberry Pi 3 version B that connect with Mininet emulator through home wireless network.

In this paper, we suggest a smart system for smart homes based on Sdn with low cost to prevent both internal and external attacks on the devices. Machine learning conducted experiments on extracted the Features related to the DDoS attack from flow status information of the network collected by the controller and apply a Random Forest classifier to distinguish the normal and abnormal traffic, then block the attack port with host. The overall accuracy rate of the experiment reached 92.2 %, The findings show that our proposed method can applicable in smart homes that provide more accuracy in identifying an IoT cyberattack in real-time.

We evaluated the performance Deep learning model of our proposed system on the dataset [27] and featured described in table II, the performance metrics were accuracy, precision, f1-score, and a confusion matrix , The best obtain result was got for binary classification using LStm with Adam optimizer with learning rate=0.001 and epsilon=1e-07 t, the overall accuracy was as shown in table III, the result of multiclassification best performance obtained using CNN with Adam optimizer as described in table II. The confusion matrix for binary classification described on Fig. 10.

TABLE III. The best Metrics performance

	Algorithm	Optimizer	Accuracy %	f1-score %
Binary Classification	LSTM	Adam	98.3	85.9
	CNN	Rmsprop	98.1	84.7
Multi calssification	LSTM	SGD	84.3	73.2
	CNN	Adam	86.1	75.8

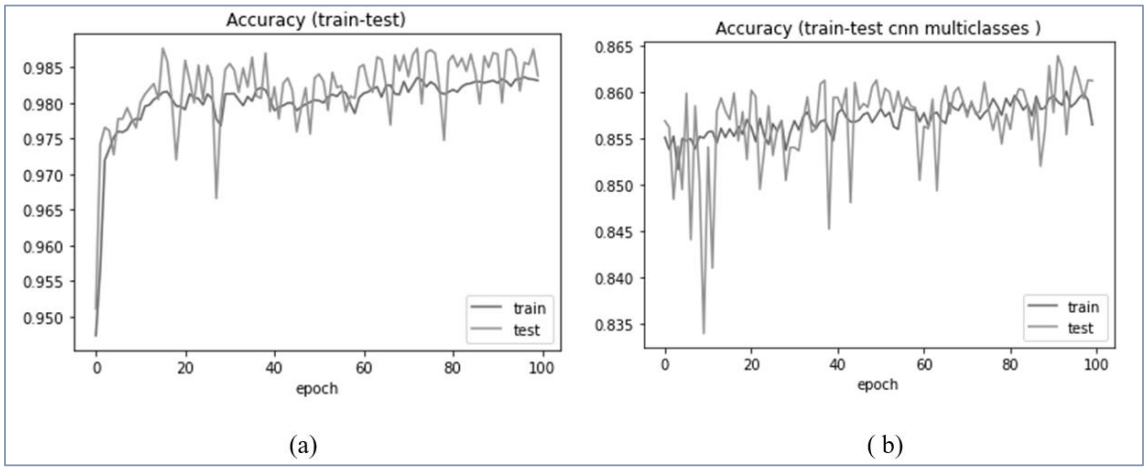


Fig. 9. (a) LSTM Accuracy for binary calssification , (b) CNN Accuracy for multicalssification

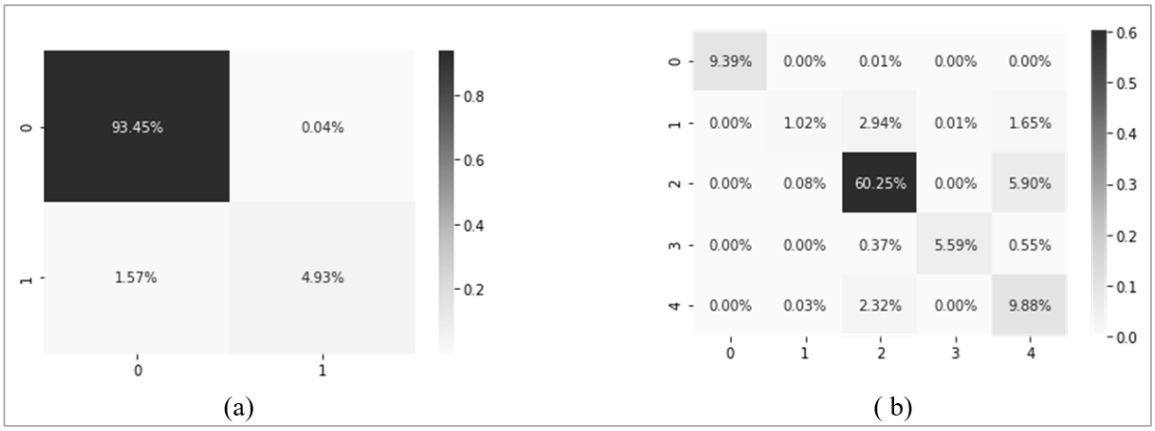


Fig. 10. (a) Confusion matrix for LSTM binary calssification, (b) Confusion matrix for CNN on multicalssification.

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VI. Conclusion

An automated preliminary, intelligent Middle cyber-attack detection and mitigation system were proposed in this paper for smart homes using SDN with deep learning. This approach relies on Zodiac-Fx SDN open flow Switch, controller install on Raspberry mini-computer and automated and second Raspberry work as gateway for IoT devices, provide authenticating IoT in the raspberry gateway provided by lightweight protocol, Classify DDoS attack types using flow feature extraction and a random forest classifier in the SDN application layer, providing accurate detection of them.

The proposed middleware was built to provide proof of concept, the secure middleware works for both Inbound as well as Outbound IoT network traffic in smart home arae. even with modern Machine learning methods, it has not been able to detect new types of IoT attacks, which poses a threat to IoT networks. In other words, we propose the use of deep learning for this issue. We implemented a deep learning-based attack detection system for multitype attack detection in smart homes based on an SDN environment. The proposed deep learning model classes with an accuracy of 98.3% for two classes and 86.1% for multiclass. our future plan will focus on building middleware that is able to identify types of attacks on IoT Networks in real-time.

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